

Progress Mechanics:

Constructing the Session Map of Tasking Momentum and Flow State

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<https://github.com/progressmechanics/progress-mechanics>

Abstract

Progress Mechanics is a psychological framework describing how **task-based momentum** develops during work sessions. The core element is the **Momentum (MAP) Drive**, the feedback cycle of motives, actions, and progress that operates across four intensity levels, or **momentum gears**, including flow. The framework also introduces the **Progress Engine**, which maps how momentum, focus, and control align during work, and **Motive Geometry**, which expands the motivation component of the momentum drive across three axes: journey vs. destination value, intrinsic vs. instrumental value, and valence. Together these models provide a structured basis for understanding and cultivating momentum in practical work contexts. This paper presents the foundational structure for a broader research program.

Keywords: tasking momentum; self-regulation; flow; progress feedback; human performance

Conceptual Core of Tasking Momentum

1. The Momentum Drive

This paper lays out a basic theory of how **tasking momentum** develops during work sessions. Momentum is often treated as a loose metaphor for motivation or productivity, when performance rises or falls. Many existing models in cognitive performance focus on motivation, self-regulation, or effort, but they usually treat these pieces in isolation. **Progress Mechanics** takes a different approach – it treats momentum as an ongoing feedback cycle with specific parts that constantly influence each other. The framework borrows language from physics, to express metaphorical forward tasking motion.

In progress mechanics, motives, actions and progress *compose* tasking momentum. The three elements are easily understood as:

- Motives (what do I want)
- Actions (what am I doing)
- Progress (how am I doing)

The core of progress mechanics is the **momentum drive**, or **MAP drive** – this is the psychological feedback loop between motives, actions and progress. As such, motives, actions and progress are defined as the three **momentum components**. Each momentum component pushes the next. Figure 1 illustrates the momentum drive, or MAP drive.

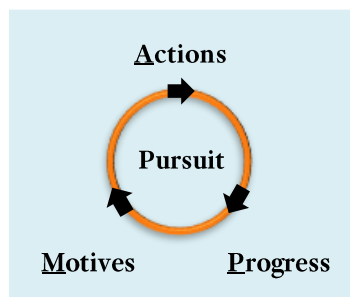


Figure 1
The Momentum Drive, or MAP Drive

Figures in this paper illustrate conceptual logic structures rather than statistical or computational models. They are intended as representational schematics of process relationships, rather than quantified systems.

1.1 The First Law of Progress

The **First Law of Progress**, or the **Law of Consistency**, states, "A task in motion tends to stay in motion, when pushed consistently." In other words, one generally must take action to keep things going. Without “pushing” a task with action, the tasking momentum or "motion" will tend to eventually slow and stop.

Such a dynamic was actually best reflected in ancient physics. A push or pull would impart motion to an object – that energy being imparted was called the impetus. It was rather like a wind-up charge. The understanding was that the object's impetus would be temporarily charged up, and then eventually ran out again if not replenished. After all, one had to keep pushing or pulling the object to keep it moving.

In progress mechanics, **impetus** is momentum requiring consistent pushing to keep moving. One can say our tasks and pursuits have an **impetus dynamic**, or are **impetus-driven**.

1.2 Session Momentum Gears

Four degrees of momentum intensity, or **momentum gears** appear across work sessions. This model builds on existing work in self-regulation and motivation (Carver & Scheier, 1998; Ryan & Deci, 2001; Csikszentmihalyi, 1990). See Table 1 for the four momentum gears:

Table 1
The four momentum gears

State	Description	Subjective feel
Cool	Idle or recovery phase.	Calm or disengaged.
Warm	Ramp-up period with early successes establishing rhythm.	Warming interest.
Hot	Higher rate of action-feedback exchange.	Energized, forward-leaning.
Flow	Self-sustaining momentum, minimal friction.	Effortless control and absorption.

The momentum gears are also illustrated in Figure 2.

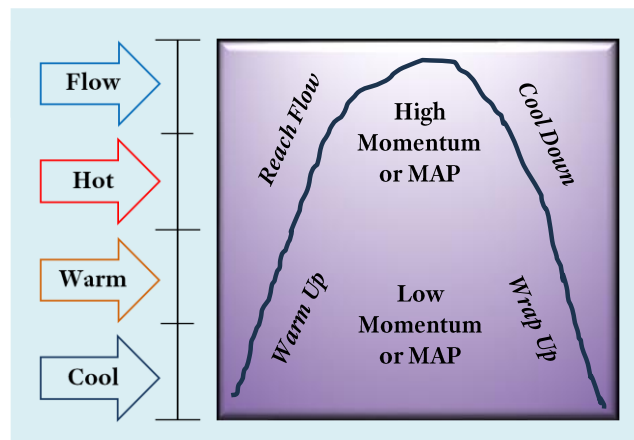


Figure 2
The Four Momentum Gears in a Work Session

With the understanding of the four momentum gears, and that tasking momentum is itself composed of motives, actions and progress, we now have the first piece of the **progress engine**. The progress engine is a blueprint for session mechanics, and is composed of three elements – momentum, focus, and control. The framework provides visual logic for sustaining motivation, self-regulation, and adaptive focus in work sessions. The progress engine, here only emphasizing momentum basics, is illustrated in Figure 3.

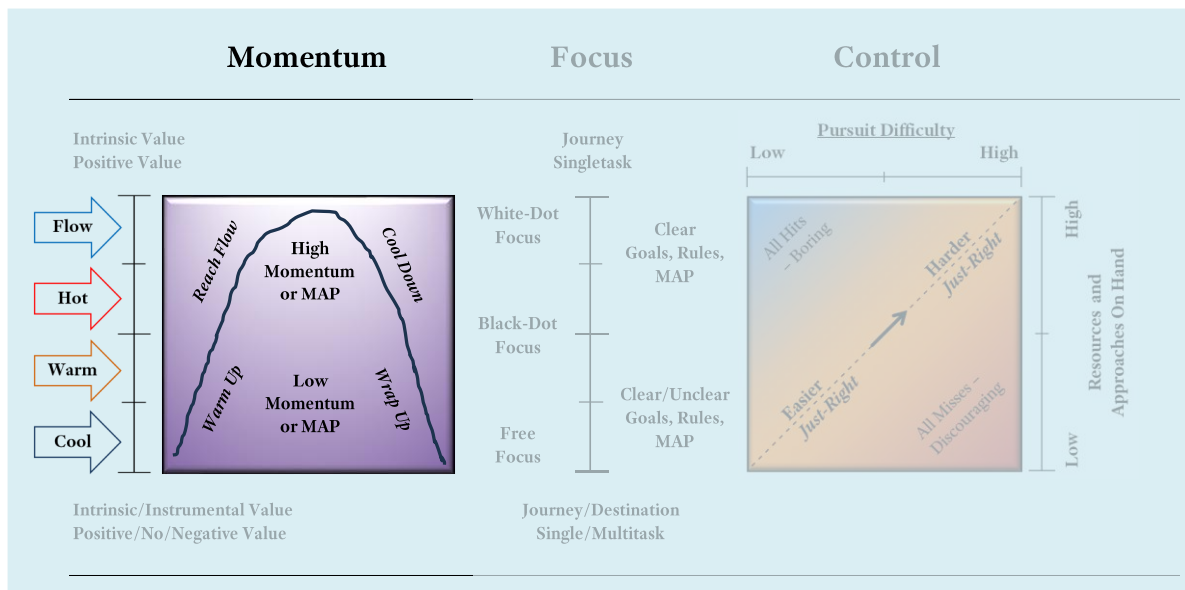


Figure 3
The Progress Engine, Momentum Basics

2. The Three Pursuit Drives: Momentum, Focus, and Control

Momentum generally depends on focus and control. Momentum, focus, and control are the three **pursuit drives**:

1. **Momentum Drive** – forward tasking motion, of motivation, action and progress.
2. **Focus Drive** – attentional direction compression, minimizing noise within the task space.
3. **Control Drive** – regulatory balance.

To make this less abstract, we can view the three pursuit drives in the context of driving:

- Momentum is moving in some direction.
- Focus is having a specific direction and destination.
- Control is steering, accelerating and braking.

These three pursuit drives are generally interdependent. When one rises or falls, the others tend to follow – the intensity of these three pursuit drives tends to align in real-time.

Figure 4 below illustrates the three pursuit drives, in basic form.

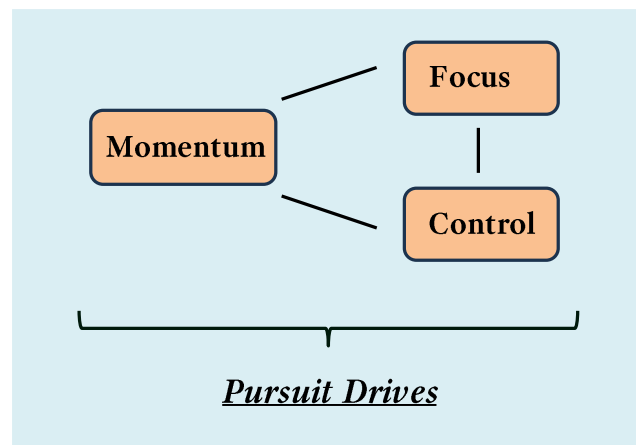


Figure 4
Pursuit Drives, Basic

The pursuit drives can be further expanded out for distinction. Motives, actions and progress compose the **momentum drive**. Focus and control are the **support drives**.

Figure 5 below illustrates the three pursuit drives in expanded form.

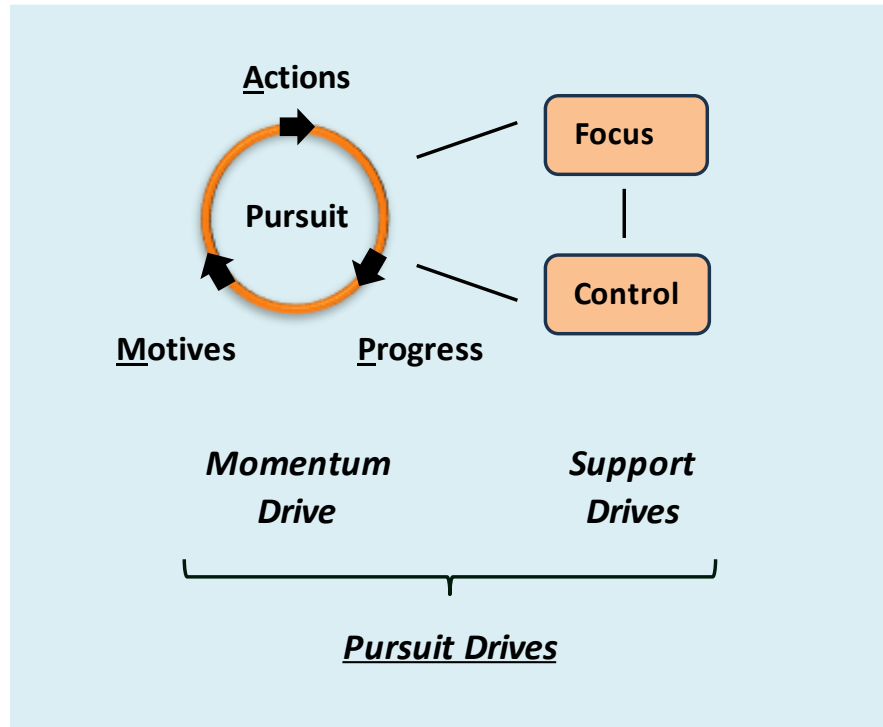


Figure 5
Pursuit Drives, Expanded

2.1 The Second Law of Progress

The **Second Law of Progress**, or the **Law of Drives** states that momentum, focus, and control tend to increase and decrease together. This is especially easy to illustrate in high-stakes contexts, where one's focus can collapse the moment control is lost – for example, a surgeon mid-complication, or a pilot in turbulence. Control provides the feedback structure that makes focus possible; focus maintains the informational economy that allows control.

3. The Focus Drive

3.1 The Three Layers of Focus

Under the progress mechanics framework, there are **three layers of focus**:

1. **Dot Focus** – the width of attentional aperture.
 - **White-Dot Focus**: singular, deep absorption.
 - **Black-Dot Focus**: fairly narrowed, possibly alternating among a few targets.
 - **Free Focus**: diffuse, exploratory.
2. **Task Number** – singletask vs multitask structure.
3. **Clarity** – precision of goals, rules, and MAP feedback.

The three layers of focus are illustrated in Figure 6 below.

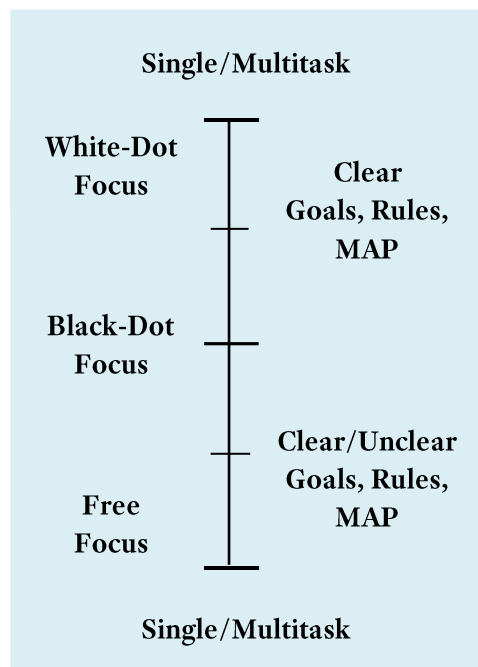


Figure 6
The Three Types of Focus

When all three focus layers narrow simultaneously – white-dot focus, singletasking focus, and clarity of focus – the system minimizes interference and the MAP drive “spins” at its highest speed.

These basics of focus provide the second piece of the progress engine, as shown in Figure 7.

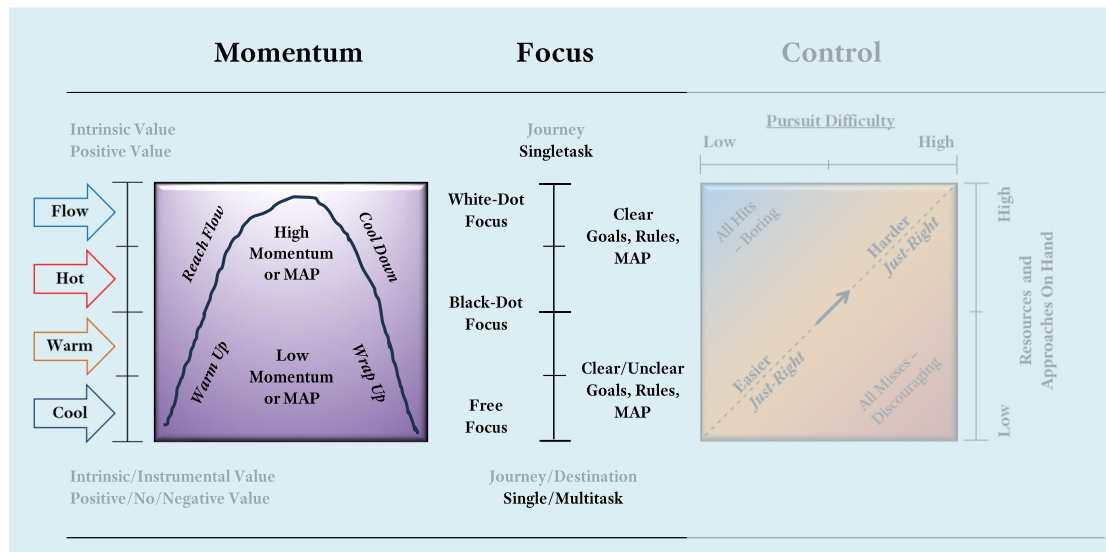


Figure 7
The Progress Engine, Momentum and Focus Basics

3.2 Clarity conditions

High-momentum sessions generally share three traits:

- **Clear goals** – destinations known.
- **Clear rules** – permissible routes defined.
- **Clear MAP** – motivations, actions, and progress are easy to perceive and understand.

Clarity of progress is based around clear, real-item feedback.

4. The Control Drive

4.1 Pursuit Balance

Under the progress mechanics framework, control in a tasking session is largely based on **pursuit balance**, which weighs difficulty against wherewithal – resources and approaches on hand. The higher the wherewithal, the higher the difficulty that can be managed.

If a task is too easy, one tends to get bored. If too hard, one tends to get discouraged. Between these lies **just-right difficulty**. See Figure 8 below for the diagram of pursuit balance.

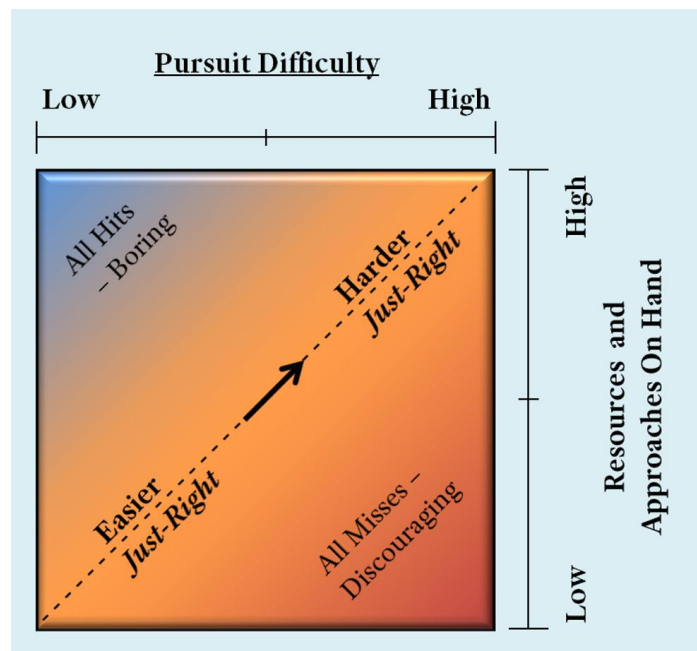


Figure 8
Pursuit Balance

Control provides the third piece of the progress engine, as shown in Figure 9 below.

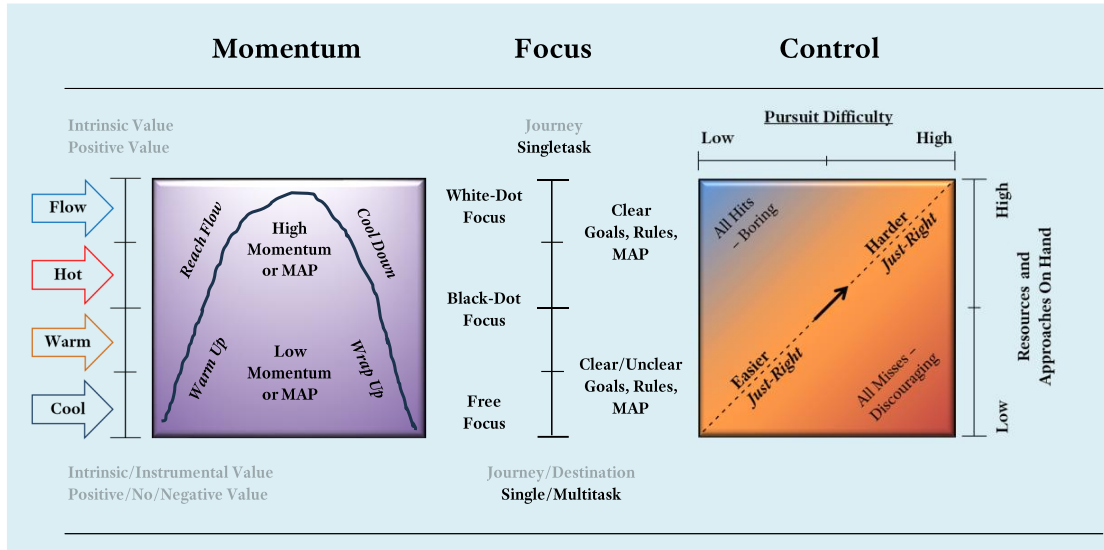


Figure 9
The Progress Engine, Momentum and Focus Basics, and Control

5. Motive Geometry

5.1 Definition

With momentum, focus, and control now defined, we can examine the motivational architecture that drives them. Under the progress mechanics framework, **motive geometry** illustrates a task's motivational profile along three **motive scales**:

1. **Journey-Destiny Scale** – the journey vs. the destination. Value in process vs. outcome.
2. **Inty-Insy Scale** – intrinsic vs. instrumental value, respectively referring to the worth of a goal for its own sake vs. as a means to an end.
3. **Valence Scale** – valence on a spectrum of three coordinates - negative (avoid), neutral (ignore), and positive (pursue).

These three axes create the **motive cube** - every task occupies at least one coordinate corner.

Tasks nearest the journey-intrinsic-positive corner generally provide the highest momentum, via support of want-to motivation and single focus. See the motive cube in Figure 10 below.

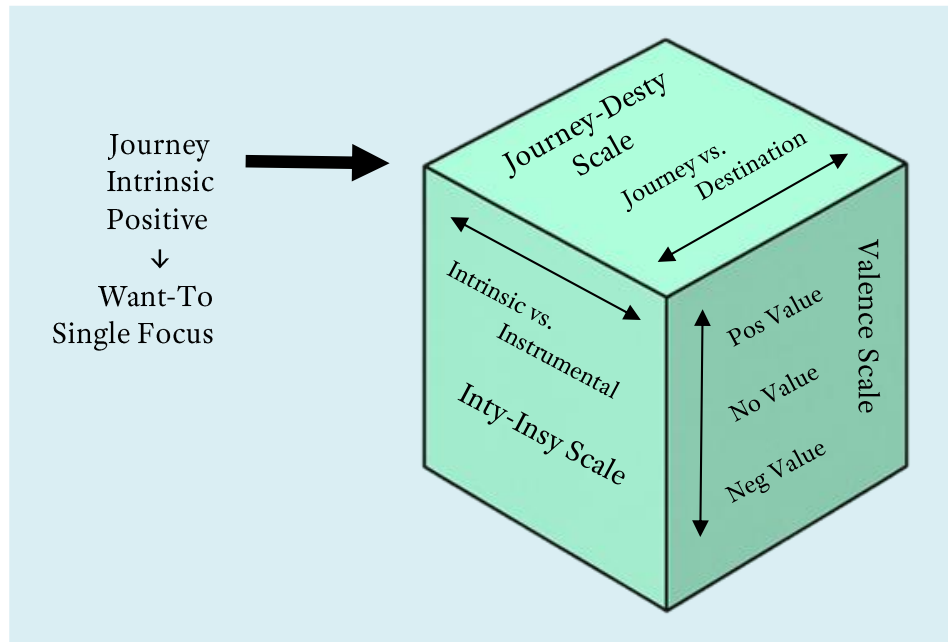


Figure 10
The Motive Cube

1. Journey / Intrinsic / Positive
2. Journey / Intrinsic / Negative
3. Journey / Instrumental / Positive
4. Journey / Instrumental / Negative
5. Destination / Intrinsic / Positive
6. Destination / Intrinsic / Negative
7. Destination / Instrumental / Positive
8. Destination / Instrumental / Negative

Each corner predicts momentum behavior. Corners with aligned positive valence accelerate the momentum drive; negative corners introduce friction.

5.2 The Motive Cube and Board

Flattening the motive cube produces the **motive board** – the 3x3 grid of possible combinations of axis values. Green zones represent aligned positive motives, yellow zones conflicted motives, and red zones aligned null and/or negative motives.

		Journey - Intrinsic Value			
		Negative Avoid	No Value Ignore	Positive Pursue	
Destination - Intrinsic Value	Positive Pursue	Avoid Journey? Pursue Destination Journey has Positive Instrumental Value Conflicted - Have-To? Split Focus?	Ignore Journey? Pursue Destination? Journey has Positive Instrumental Value Conflicted - Have-To? Split Focus?	Pursue Journey Pursue Destination Journey has Positive Instrumental Value Aligned - Want-To Split Focus?	1
	No Value Ignore	Avoid Journey Ignore Destination Aligned - Avoid No Focus	Ignore Journey Ignore Destination Aligned - Ignore No Focus	Pursue Journey Ignore Destination Aligned - Want-To? Single Focus	2
	Negative Avoid	Avoid Journey Avoid Destination Journey has Negative Instrumental Value Aligned - Avoid No Focus	Ignore Journey Avoid Destination Journey has Negative Instrumental Value Aligned - Avoid No Focus	Pursue Journey Avoid Destination Journey has Negative Instrumental Value Conflicted - Want-To? Split Focus?	3
		C	B	A	

Figure 11
The Motive Board

- **Green zones** (A1, A2): both journey and destination positive → want-to.
- **Yellow zones** (C1, B1, A3): mixed values → internal conflict.
- **Red zones** (C2, B2, C3, B3): negative and/or null → disengagement.

Momentum thrives in green zones, is conflicted in yellow, and will not proceed in red. A weak but aligned motive (modest enjoyment + mild reward) can outperform a strong but conflicted one (excitement towards a dreaded end).

5.3 Conflict triangle

The **motive conflict triangle** (or **motive triangle** or **conflict triangle**) is visual logic expressing the four conflicted yellow zones on the motive board. This gives us four types of motivational conflicts, or trade-offs. The premise is to start the tasking journey in one triangle corner, and reach the second corner as the destination, in a trade-off – see Figure 12 below.

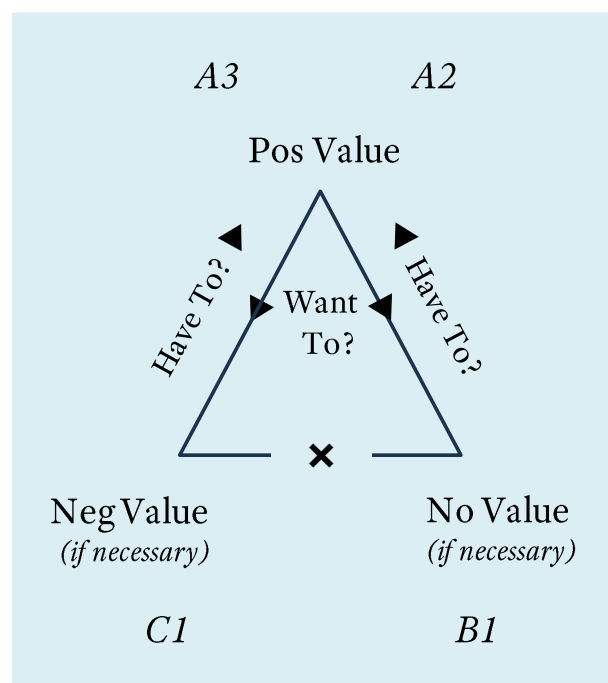


Figure 12
The Motive Conflict Triangle

- On the top corner, we start the journey with positive intrinsic value. The task is a want-to. However the destination has negative value (square A3) or no value (square A2). With A3, there is a conflict of interest – one can have fun now, and pay the consequences later. With A2, there *may* be a conflict of interest, if the lack of payoff is actually an issue.

- On the left bottom corner, we start with the journey having negative intrinsic value.

However that journey leads to a destination with positive value. The journey is a have-to – it is merely transactional. This conflict of interest matches square C1.

- On the right bottom corner, we start with the journey having no intrinsic value.

However that journey leads to a destination with positive value. The journey is a have-to – it is merely transactional. This conflict of interest matches square B1.

Squares C1 and B1 do not lead to each other in practical terms – neither motivates pursuit of a task at hand. There would therefore be no reason to pursue a tasking journey or destination.

One can have many motivations attached to a single goal. Some motivations may align, while others conflict. There may even be many motivations fitting within a single board square. One may find themselves overwhelmed and undecided. Ultimately it might make sense to create a list of pros and cons, to decide the best course of action.

5.4 Auto Euporia: The Self-Sustaining State

At the upper left corner of the motive cube lies **auto euporia** – the motivational state of intrinsic, positive value of the journey. Auto euporia derives from Greek roots *auto* (self, or intrinsic), *eu* (good, or positive), and *poria* (from *poreia* as journey). Auto euporia matters because it supports want-to focus and single tasking, which in turn aligns with the higher momentum gears. See the motive cube with auto euporia in Figure 13 below.

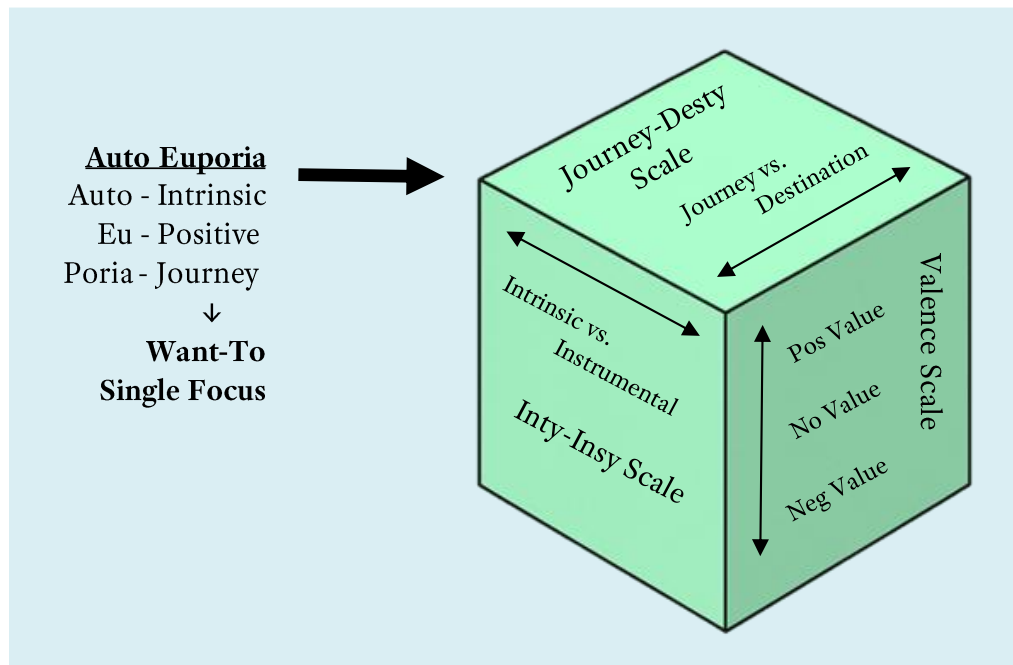


Figure 13
The Motive Cube with Auto Euporia

In the auto-euporic state, motivational focus is based on want-to's (aligned), rather than have-to's (conflicted). Motivational focus also leans towards having a single focus rather than a split focus. Motivational alignment and singletasking increase focus, in turn likely increasing momentum, as per the Second Law of Progress.

This overall dynamic is illustrated in Figure 14, below.

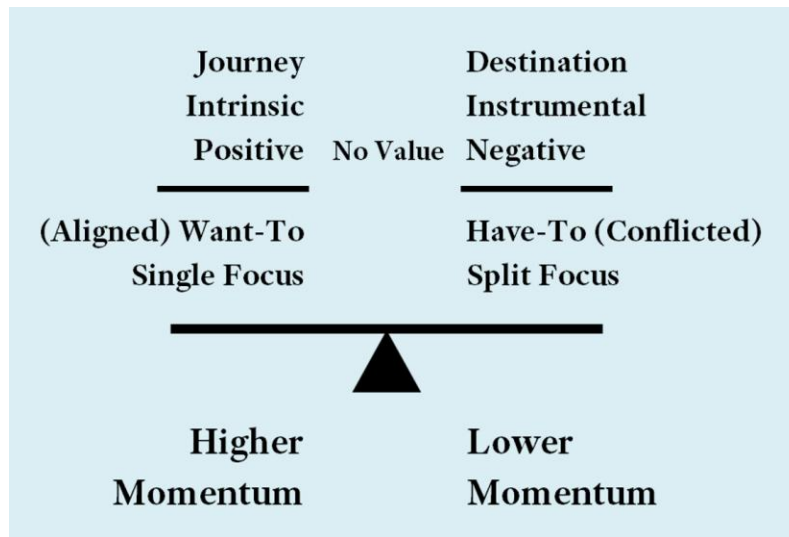


Figure 14

Alignment of the Motive Cube Axes with Higher and Lower Momentum

6. The Completed Progress Engine

With auto euphoria, the progress engine is now complete. As shown, intrinsic and instrumental value, and valence are established under momentum. Motivational focus of journey and destination is based under Focus. Auto-euporic motivation - positive, intrinsic value of the journey – is shown on top to align with higher momentum. See the completed progress engine, as the blueprint of session momentum, in Figure 15 below.

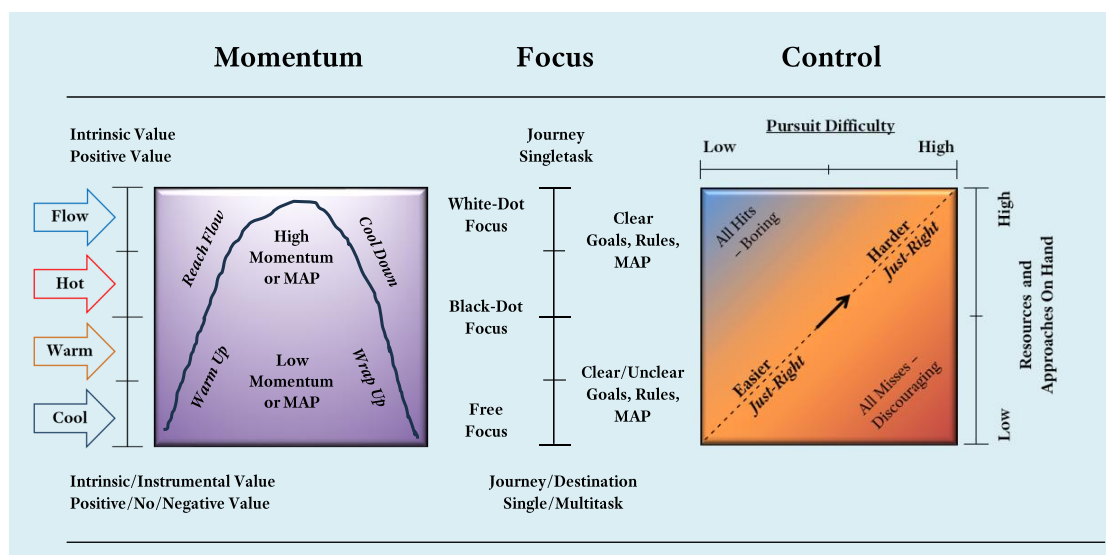


Figure 15
The Progress Engine, Fully Clarified

6.1 Flow as Mechanical Synchronization

When auto euphoria coincides with momentum, focus, and control from pursuit balance, the system can more easily achieve higher momentum. These are the psychological conditions required to produce the classic **flow state** (Csikszentmihalyi, 1990). Within the progress engine, each of the nine phenomenological characteristics of flow corresponds to a mechanical property of the MAP drive:

1. **Clear goals** → established under MAP clarity, under clarity of motivation.
2. **Immediate feedback** → established under MAP clarity, under clarity of progress.
3. **Balance between challenge and skill** → Higher pursuit balance under control corresponds with higher momentum and focus, as per the Second Law of Progress.
4. **Merging of action and awareness** → in the higher momentum gears, the MAP drive of motives, actions, and progress can “spin” faster than conscious reflection can interrupt it.
5. **Concentration on the task at hand** → maximal focus compression (white-dot focus).
6. **Sense of control** → stability of control drive; neither chaos nor rigidity.
7. **Loss of self-consciousness** → reduced self-monitoring is an *effect* of higher focus, as energy is devoted entirely to tasking.
8. **Transformation of time** → subjective time dilation or constriction is an *effect* of reaching a more focused cognitive state.
9. **Autotelic experience** → this comes of sustained auto euphoria – positive, intrinsic value of the tasking journey.

Under the progress engine framework, the flow state is illustrated as the highest momentum gear, supported by highest focus, and control.

7. Conclusion: The Architecture of Human Motion

In this paper, the foundation of progress mechanics is introduced. The progress engine is a theoretical blueprint of session mechanics, based on the interplay between the three pursuit drives – momentum, focus, and control. Momentum is tasking motion; focus provides direction; control provides steering, accelerating, and braking. In general, the intensity of these three pursuit drives aligns in real-time.

Momentum is based on the momentum drive, or MAP drive – motives, actions and progress – as a self-reinforcing drivetrain.

Motivation in turn is based on motive geometry. Motive geometry is composed of the motive cube, motive board, and motive conflict triangle. These provide visual frameworks for tasking motivation, leading up to auto euporia (auto-euporic motivation), or positive, intrinsic value of the journey.

When momentum, focus, and control align at high intensity – the progress engine reaches the optimal flow state. This is not mystical immersion but mechanical resonance. The task and self can even become indistinguishable because the feedback delay approaches zero; every act immediately verifies its own purpose.

In essence:

Progress is not a mystery to be chased, but a workable mechanism to be mastered.

The progress mechanics framework discussed in this paper represents the foundation of a much broader research project. A book-length treatment is in preparation and will provide extended analysis, examples, and applied methods.

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